

Business Requirement

In today's digital music era, understanding listening patterns is crucial for both users and streaming platforms. This analysis focuses on Spotify Albums Data, providing insights into user engagement with albums over time.

ALBUMS

- 🎵 **Total Albums Played Over Time** – Track how album listening trends change over months and years.
- 📅 **Number of Albums Listened by Year** – Identify annual listening habits and volume (Find the Min and Max Albums in the view).
- ⭐ **Albums Played on Weekday & Weekend** – Identify the Pattern of music listening on weekdays and weekends.
- 🏆 **Top 5 Albums** – Identify the most played albums based on listening frequency.
- 📊 **Latest Year vs Previous Year Analysis** – Compare album consumption between the latest and previous years, including:
 - ⚡ **LY (Latest Year) vs PY (Previous Year) Trends**
 - ⚡ **YoY (Year-over-Year) Growth Analysis**

ARTISTS

- 🎵 **Total Artists Played Over Time** – Track how artist listening trends evolve across months and years.
- 📅 **Number of Artists Listened by Year** – Identify annual listening habits and artist diversity. (Find the Min and Max Artists in the view).
- ⭐ **Artists Played on Weekday & Weekend** – Identify the Pattern of music listening on weekdays and weekends.
- 🏆 **Top 5 Artists** – Identify the most played artists based on listening frequency.
- 📊 **Latest Year vs Previous Year Analysis** – Compare artist engagement between the latest and previous years, including:
 - ⚡ **LY (Latest Year) vs PY (Previous Year) Trends**
 - ⚡ **YoY (Year-over-Year) Growth Analysis**

TRACKS

- 🎵 **Total Tracks Played Over Time** – Monitor how track listening trends change across months and years
- 📅 **Number of Tracks Listened by Year** – Identify annual listening habits and track diversity. (Find the Min and Max Tracks in the view).
- ⭐ **Tracks Played on Weekday & Weekend** – Identify the Pattern of music listening on weekdays and weekends.
- 🏆 **Top 5 Tracks** – Identify the most played tracks based on listening frequency.
- 📊 **Latest Year vs Previous Year Analysis** – Compare track engagement between the latest and previous years, including:
 - ⬆ **LY (Latest Year) vs PY (Previous Year) Trends**
 - ⬆ **YoY (Year-over-Year) Growth Analysis**

LISTENING PATTERNS

- 🕒 **Listening Hours Analysis** – Identify peak listening times using a **Heat Map** that visualizes patterns across hours and days with color intensity.
- 📊 **Average Listening Time (min) vs Track Frequency** – Use a **Scatter Plot with Quadrant Analysis** to categorize tracks based on:
 - ⬆ **High Frequency & High Listening Time** – Most engaging tracks 🎯
 - ⬆ **Low Frequency & High Listening Time** – Niche but impactful tracks
 - ⬆ **High Frequency & Low Listening Time** – Short & frequently played tracks
 - ⬆ **Low Frequency & Low Listening Time** – Less popular tracks

DETAILS GRID

In this report, we aim to analyze Spotify data by creating an interactive and dynamic **Grid View**. The Grid will display key details such as **Album Name**, **Artist Name**, **Track Name**, and other relevant attributes.

Key Requirements:

1. Grid View with Essential Fields:

1. The Grid should present critical data points for an intuitive and structured view.

2. Drill Through Functionality:

2. Users should be able to drill through from the main reports to explore underlying data for detailed insights.

3. The drilled-through data should be exportable to a CSV file based on user requirements.
3. **Drill Down, Drill Up, and Hierarchy:**
 4. The Grid should support hierarchical navigation, allowing users to drill down and up for in-depth data exploration.

Deliverables

1. Data Cleaning & Quality Check:

- Standardized datasets via Power Query with verified data types (Datetime, Text, Whole Number, Duration/Seconds).
- Removal of duplicates (Remove Duplicates) and handling of missing/invalid values (Replace/Remove Null/Errors).
- Filtration logic to exclude invalid or skipped streams (e.g., streams where ms_played < 30000 ms).

2. Data Modeling & Connection: Dedicated calendar table built with DAX/Power Query (Date, Year, Month, Month_Name, Day_of_Week, Is_Weekend). Master lookup tables for track, artist, and album metadata.

3. Data Processing (Calculated Columns): Quadrant Category = Quadrant matrix classification based on playback frequency thresholds and average listening duration.

4. DAX Calculations:

- Number of Albums
- Latest Year
- Previous Year Albums
- Previous Year and Year-on-Year Albums KPI

5. Dashboard Development, Visuals & Reporting:

- **Overview / KPI Header:** Summary cards displaying Total Tracks, Artists, Albums, Total Listening Time, and YoY growth percentages.
- **Albums / Artists / Tracks Breakdown Pages:**
 - Line & Area charts for month-over-month and year-over-year stream trends.
 - Donut & Bar charts showing Weekday vs Weekend distributions.
 - Ranked horizontal bar charts for Top 5 Albums, Artists, and Tracks.
- **Listening Patterns Page:**
 - **Heat Map Matrix:** Color-intensity visual mapping peak listening hours across days of the week.
 - **Scatter Plot (Quadrant Analysis):** Four-quadrant categorization plotting Track Frequency against Average Listening Time.

- **Details Grid & Drill-Through:**

- Multi-level hierarchy table with drill-down capabilities (Artist → Album → Track).
- Configured **Drill-Through** navigation from high-level visuals to raw transactional data, complete with CSV export support.

6. Insights Generation:

- Executive summary document detailing peak listening timeframes, annual growth trends, and highest-performing audio segments.
- Actionable recommendations for playlist curation, release timing, and targeted content recommendation strategies.